# Kubernetes Basics

## Warsaw University of Technology




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# Agenda

Why Kubernetes?

Architecture

Concepts

Basic Controllers

Services

Storage

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# What is Kubernetes?

Open Source platform for containerized workloads and microservices.

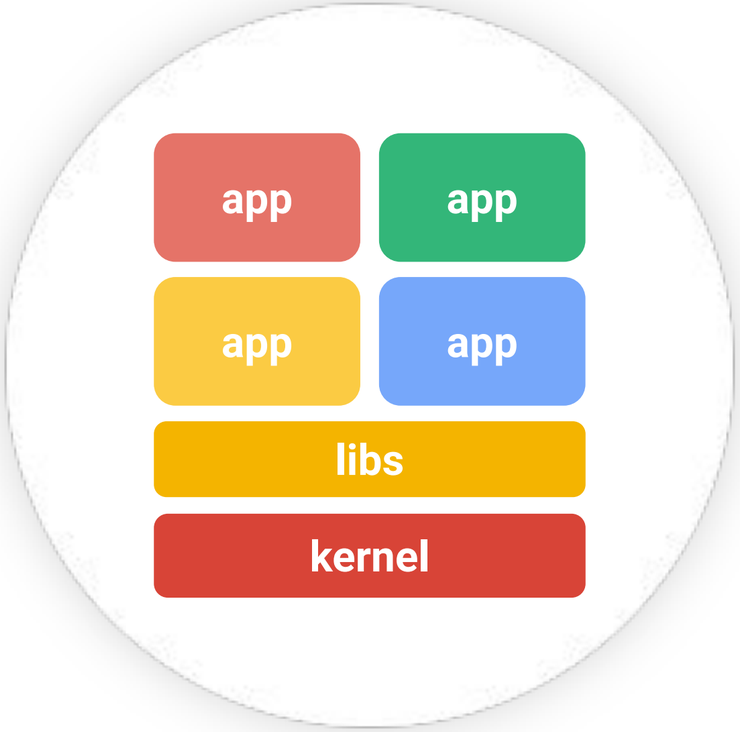
- **declarative** configuration
- portable, extensible, automated
- with thriving **ecosystem**

Based on Google experience in running Borg for 15+ years.

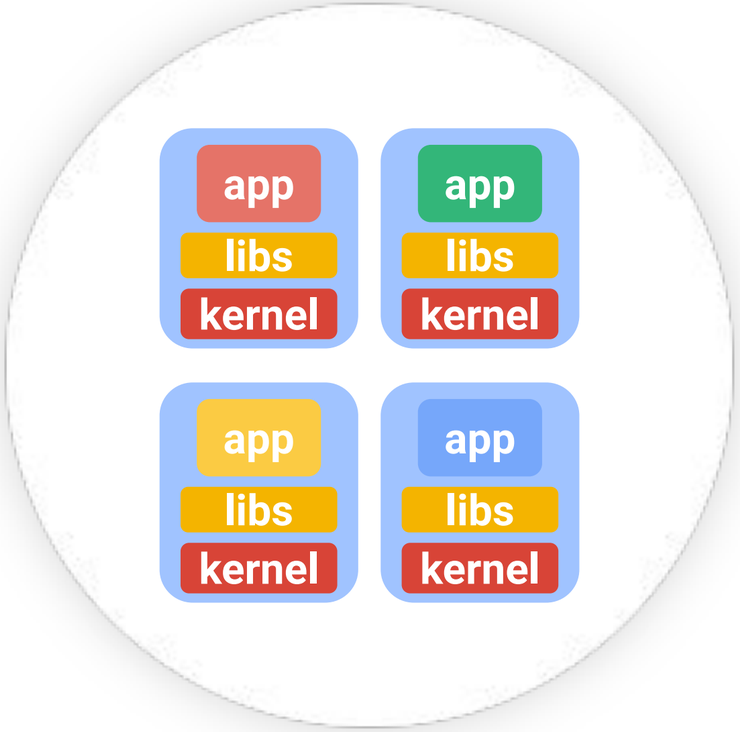


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# Road To Containers




Shared machines

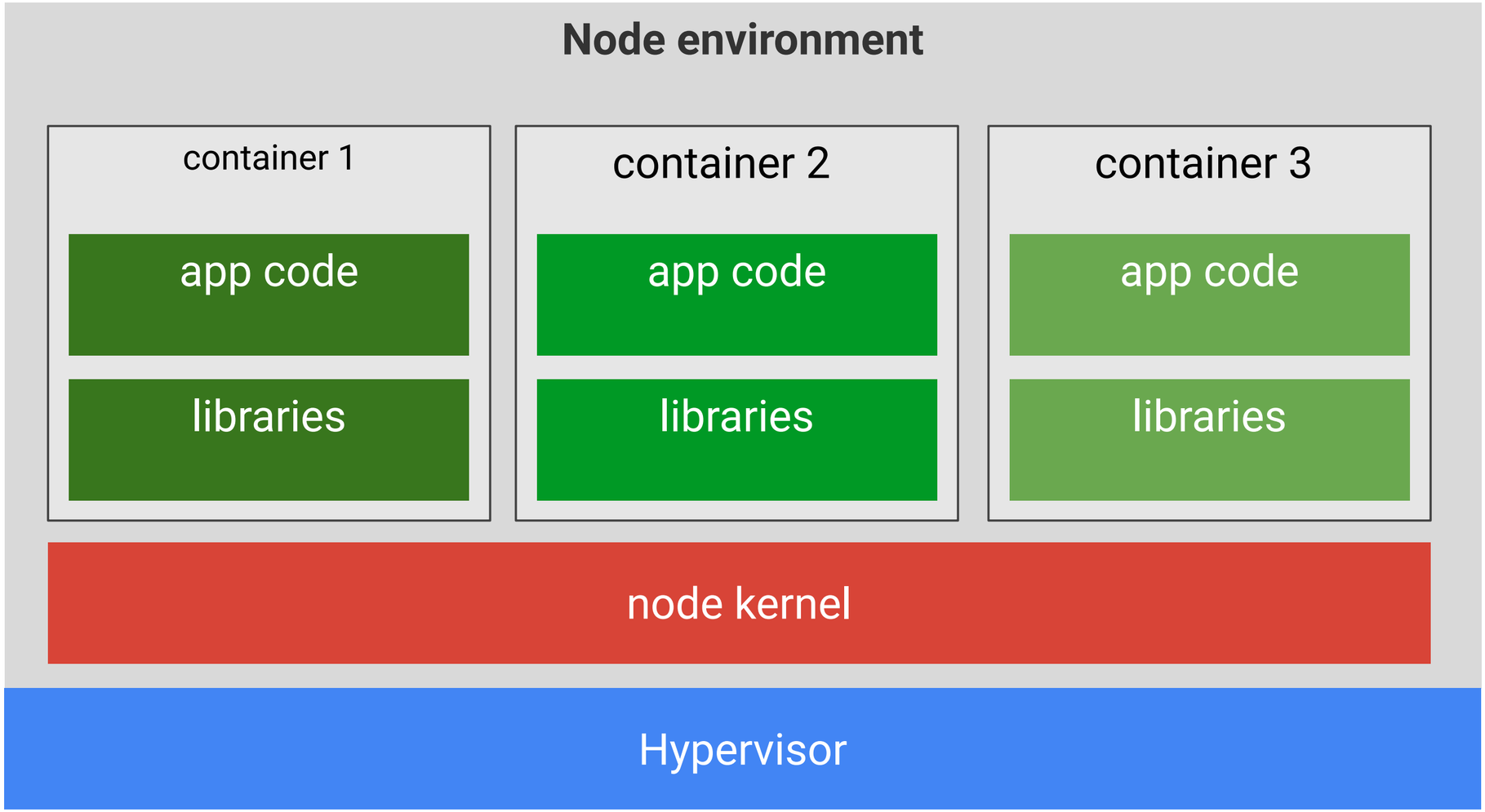



VMs/Bare metal

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Google Cloud

# Containers




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# Why Containers?

- loosely coupled, **distributed** microservices
- agile & **lightweight**
- resource **isolation**
- environment **consistency**
- continuous development, integration & deployment
- separates Devs and Ops
- observability
- portability

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# Why Kubernetes?

The **winning** platform for containers.

- portable:
  - runs in cloud or on-prem,
  - no vendor lock-in,
- thriving **ecosystem**:
  - many people,
  - many learning resources,
  - many products.



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# Why Google Kubernetes Engine?

**Managed** Kubernetes offering:

- automatic security fixes,
- taking care of **upgrades**,
- fully **managed** control plane,
- GKE Autopilot: also fully managed nodes.



Great if you want to:

- **learn** Kubernetes,
- run Kubernetes **without hassle**.

Not good if you want to **tinker** a lot.

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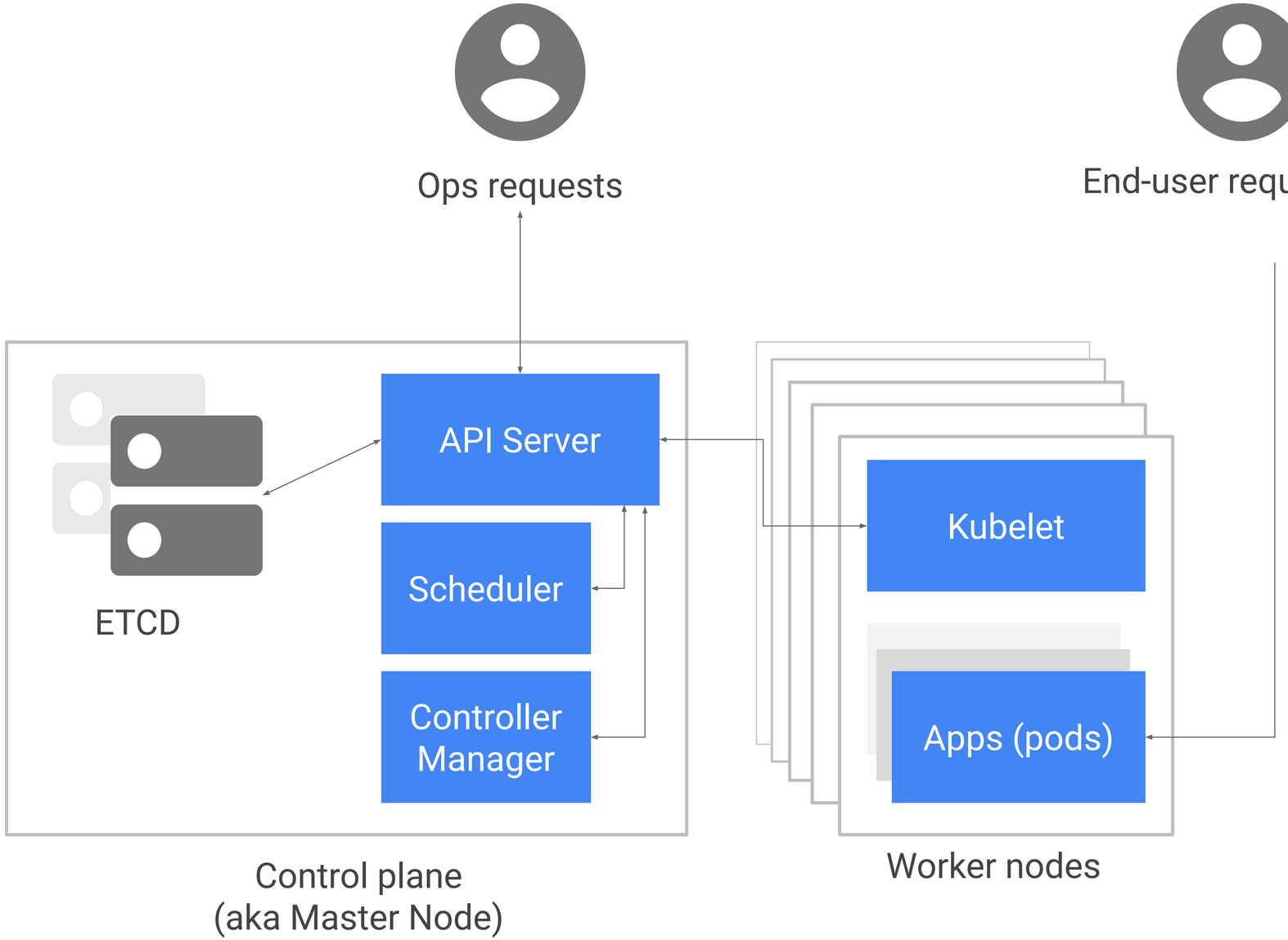
Basic Controllers

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# Architecture Overview




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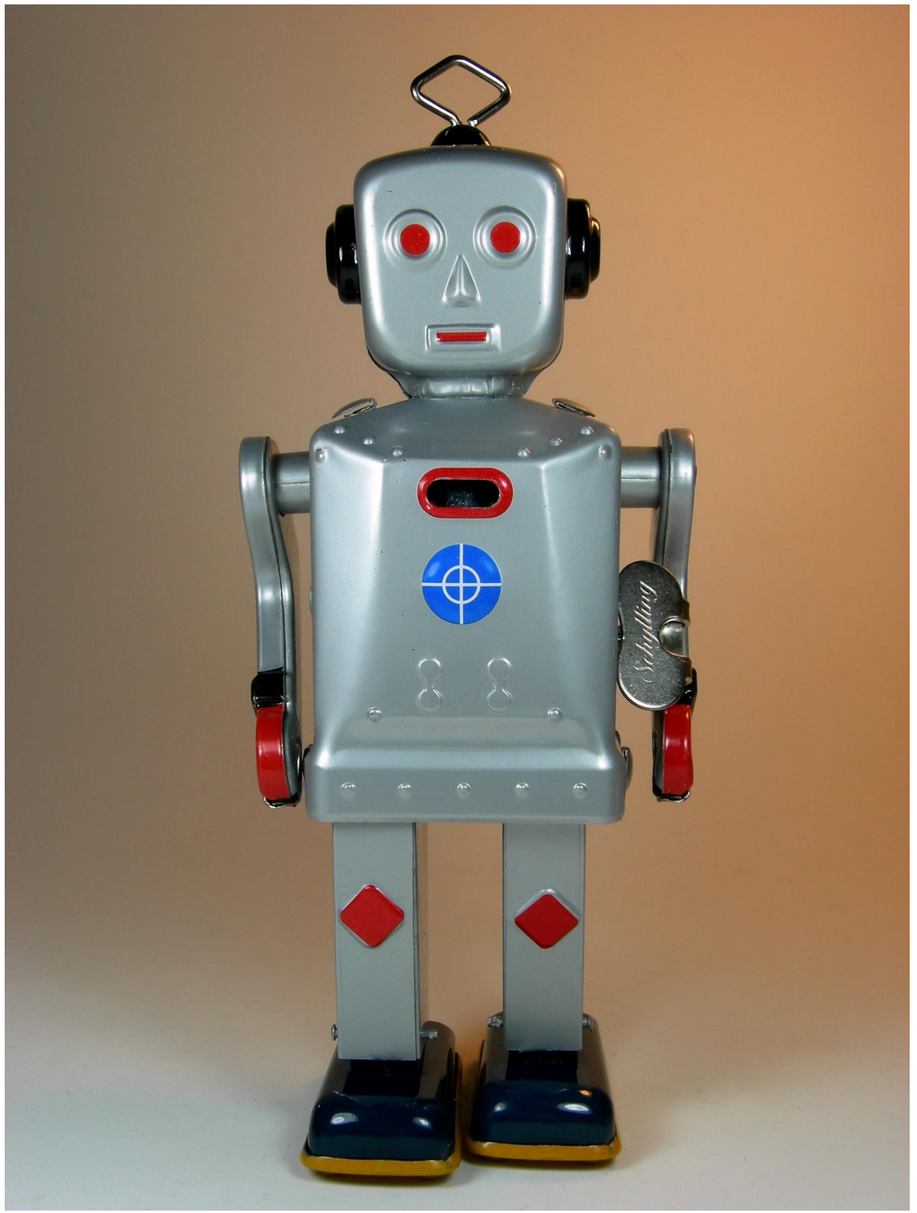
# Kubernetes Master

**Control plane** for a single cluster.

Logical entity, usually also a physical one.

- kube-apiserver
- etcd
- kube-scheduler
- kube-controller-manager
- (optional) cloud-controller-manager

GKE **regional** masters for HA.



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# Node

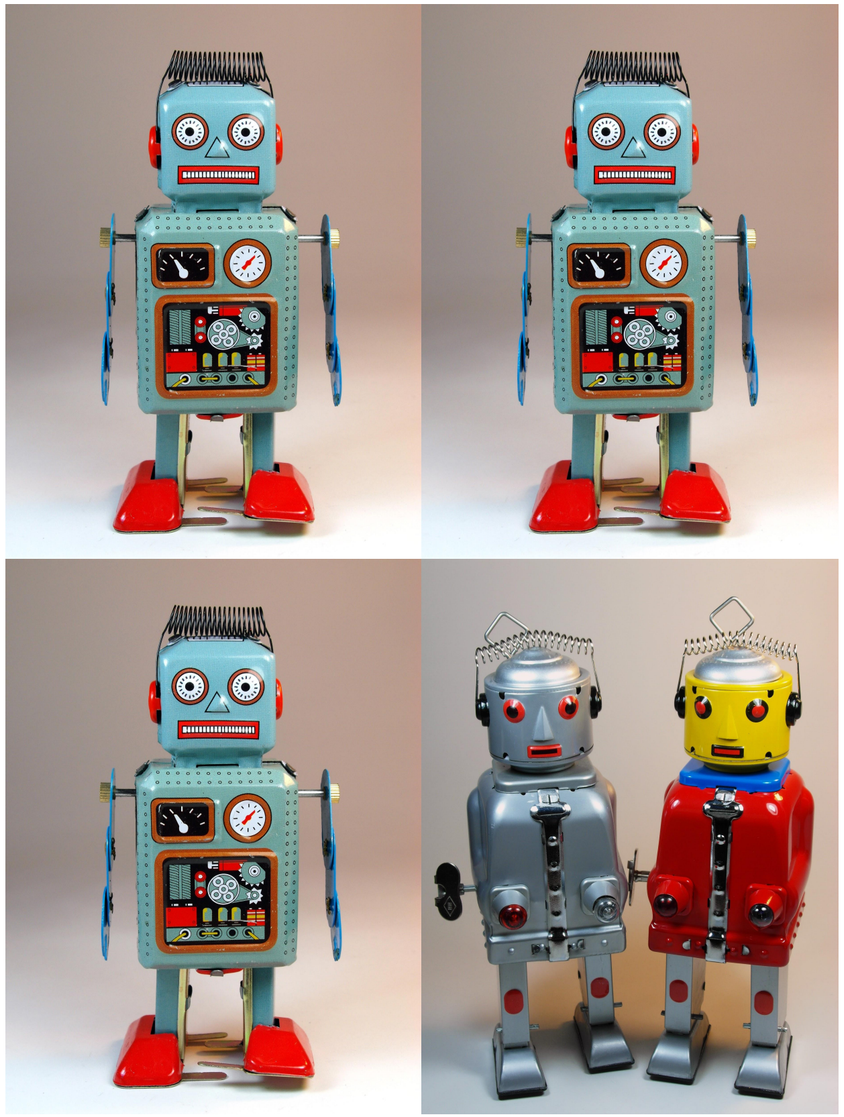
A **worker** machine, **registered** but not necessarily created.

Runs multiple user workloads and bits of control plane:

- kubelet,
- kube-proxy,
- container runtime (docker, containerd, others).

Default properties (labels):

- hostname,
- zone & region,
- instance-type, os & arch.



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# Node, cntd.

Might be created by cloud-controller-manager.

Addresses:

- hostname
- externalIP
- internalIP

**Capacity**:

- CPU
- memory
- ephemeral storage
- number of pods

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# [Reference] Node, cntd.

Conditions

- Ready
- warnings
  - MemoryPressure
  - PIDPressure
  - DiskPressure
- errors
  - OutOfDisk
  - NetworkUnavailable

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# Kubernetes Addons

Run on **Nodes**, together with user workloads.

- CoreDNS
  - automatically installed,
  - automatically included by all containers.



[Reference] Optional addons:

- Web UI (Dashboard),
- Container Resource Monitoring,
- Cluster-level Logging,
- network-related.

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# Ecosystem



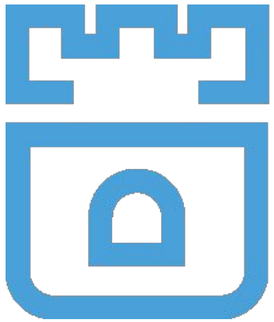















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Concepts - API

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# Kubernetes API

Common API used **everywhere**:

- no public vs private API,
- all components use the same API,
- local & remote access,
- RESTful, usually sent in JSON,
- usually shown in YAML.



Everything is an **object**:

- declarative configuration,
- stored state (in etcd),
- split by groups & versions
  /apis/extensions/v1beta1

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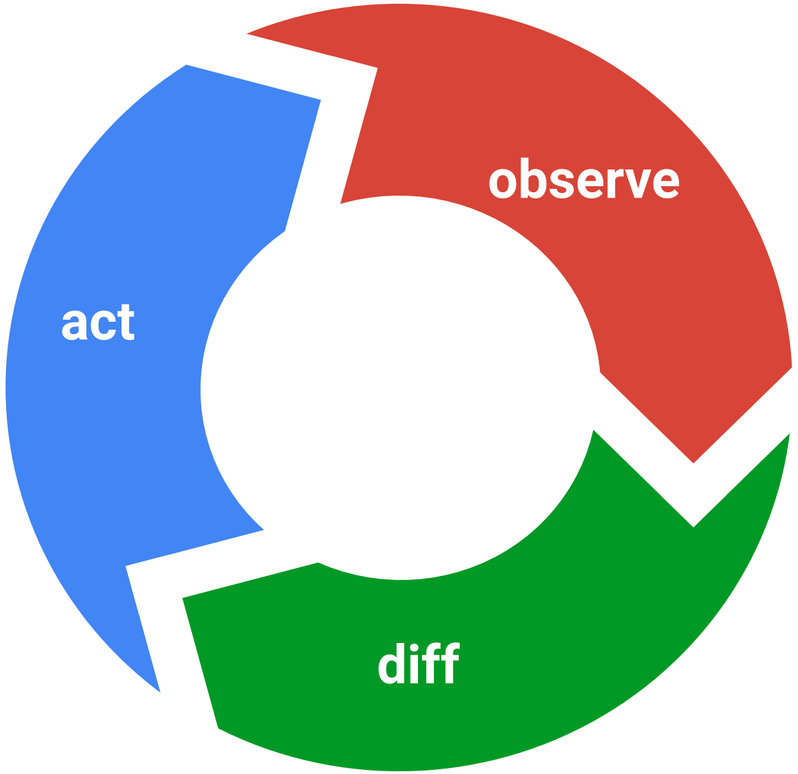
# Kubernetes API, cntd.

API is **extensible**:

- via CustomResourceDefinition objects,
- or via own apiserver and aggregation via kube-apiserver.

Different ways to interact:

- imperative commands: CRUD on objects,
- imperative configuration: CRUD on files,
- declarative configuration: diff/apply files.




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# Kubernetes API, Verbs

Extended REST pattern:

- GET - list or single object,
- POST - create new object,
- DELETE,
- PUT - update or create,
- PATCH - **selectively** modify,
- GET ...&watch=true - receive stream of updates (**WATCH**).

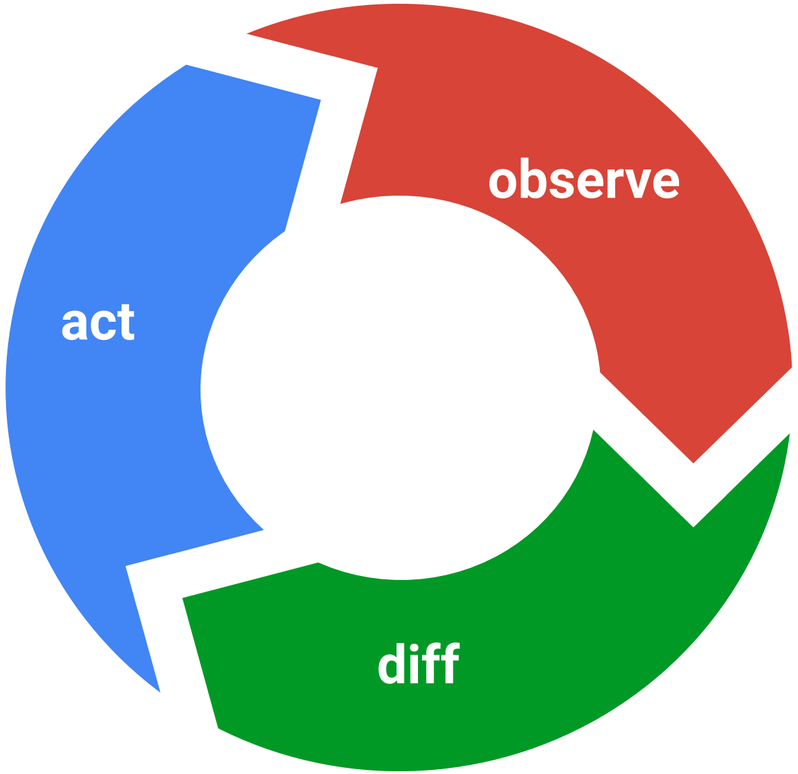
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# API Objects

- Spec - the **desired** object state.
- Status - **current**, observed state.

Required fields:

- kind
- apiVersion
- metadata
  - name
  - (defaulted) namespace
  - (generated) UID




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# [Reference] Metadata

Name

Namespace

UID

ResourceVersion

Generation

CreationTimestamp

DeletionTimestamp

Labels

Annotations

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# Labels

Dictionary of key/value pairs:

- **meaning** for users,
- acted upon by system,
- add dimensions,
- used in **selectors**:
  - equality (=, !=),
  - set (in, notin),
- used in LIST and WATCH filtering.

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# [Reference] Suggested Labels

Common labels, prefixed with app.kubernetes.io/

- name
- instance
- version
- component
- part-of
- managed-by

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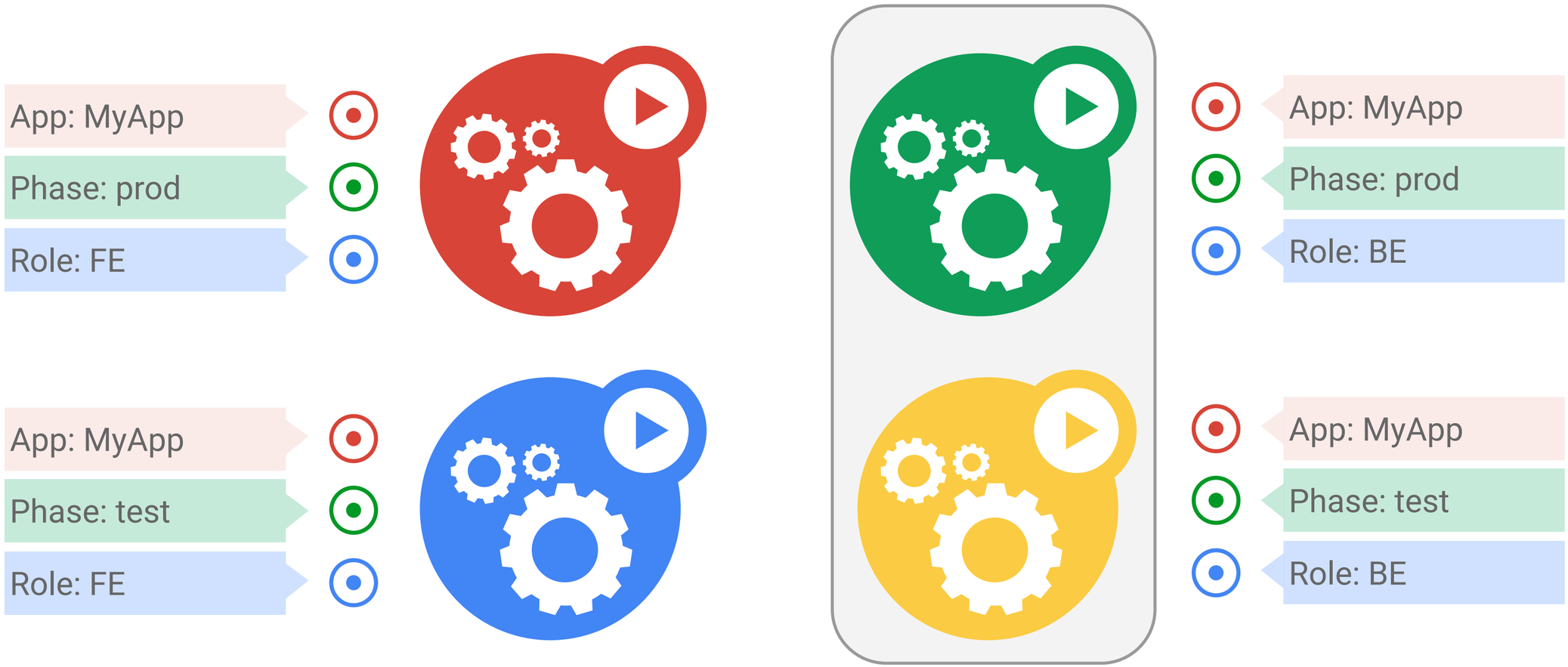
# Selectors

- Label selectors
  - equality (=, !=)
  - set (in, notin)
- Field selectors
  - name, namespace
  - others depending on resource kind

- Node selectors
  - field in PodSpec
  - standard node labels: hostname, zone, region, instance-type, os, arch

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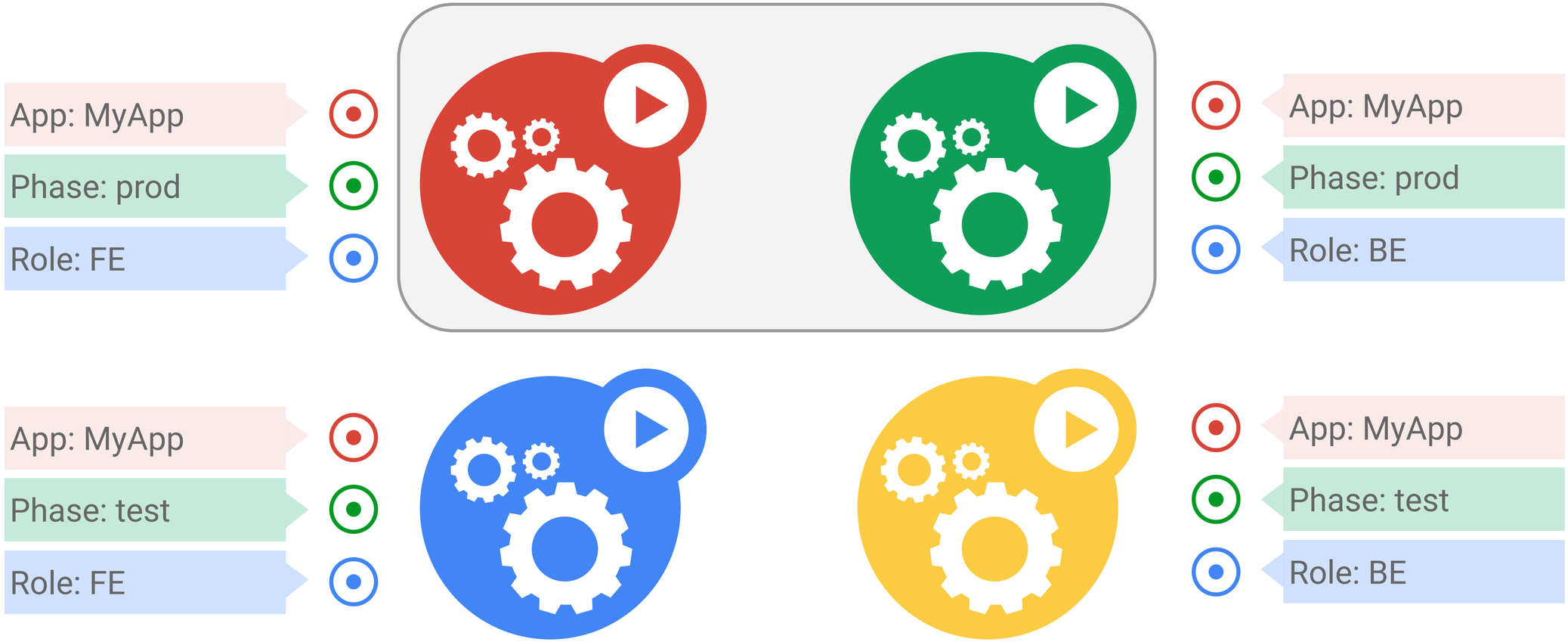
# Selectors At Work




App = MyApp, Role = BE

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# Selectors At Work




App = MyApp, Phase = Prod

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# Annotations

Dictionary of key/value pairs:

- meaning for people,
- used by external tools (in/out),
- usually **informative**, not declarative,
- values are more free-form.

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# Namespaces

Split the cluster into multiple logical clusters:

- separation of scopes,
- allow resource quota.

Not everything lives in a namespace:

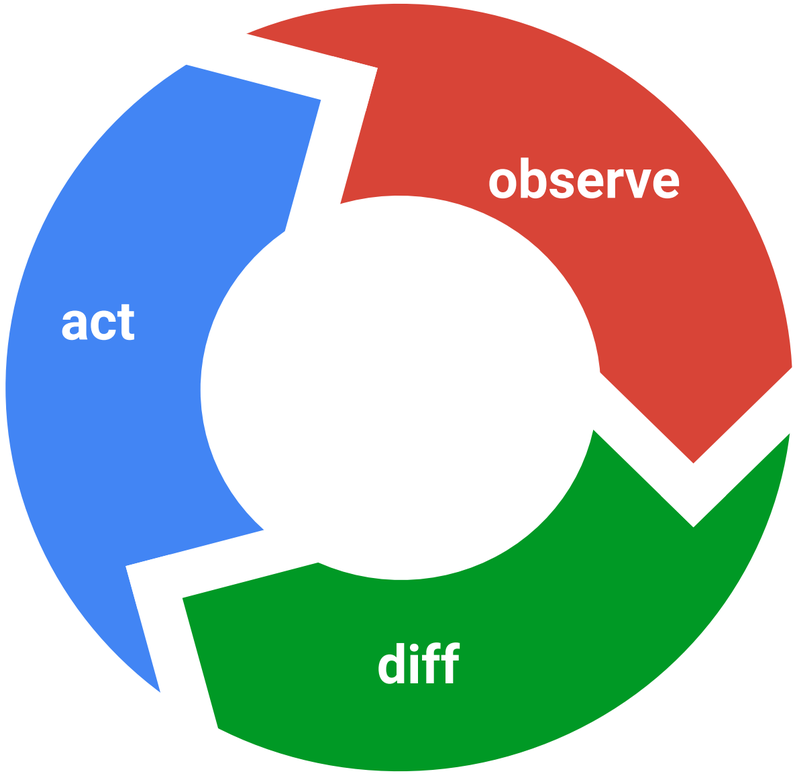
- namespaces
- low-level resources (nodes, persistentVolumes)

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# Custom Resources

- CustomResourceDefinition
  - Kubernetes object
  - **extend** Kubernetes installation
  - make Kubernetes core **modular**
  - **structured** storage
  - validation (schema & webhooks)

- Operator Pattern
  - combine with **custom controller**
  - run **control loop**




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Concepts - Workloads

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# Container

**Immutable** image

- registry or pre-loaded
  - gcr.io is auto-configured in GCE & GKE
- ephemeral local storage
- life-cycle hooks (PostStart, PreStop)
- different runtimes (RuntimeClass)

Requires **resources.**

Can have resource limits.




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# Pod

Basic **building block** for workloads:

- **immutable** PodSpec,
- single unit of work,
- one or more containers,
- **unique** IP address,
- shared storage volumes,
- ephemeral.



Runs on a single Node until deleted.

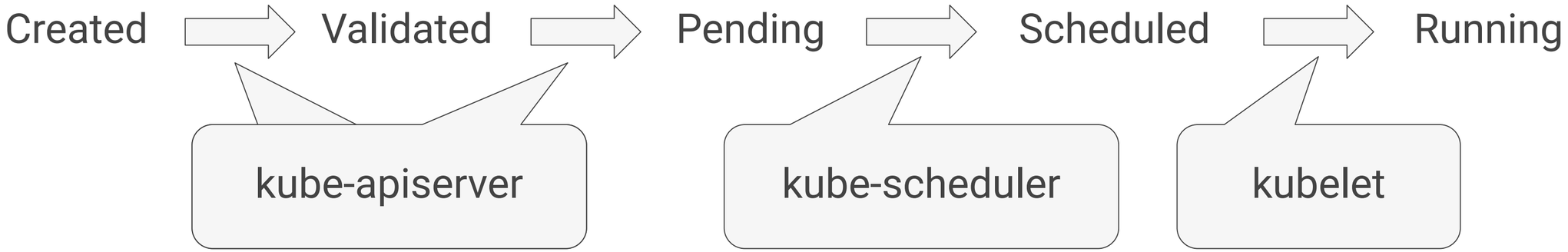
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# A Minimal Pod

```
apiVersion: v1
kind: Pod
metadata:
  name: minimal-pod
  labels:
    app: minimal
spec:
  containers:
  - name: sample-container
    image: ubuntu-slim
    command: ['sh', '-c', 'echo Hello Kubernetes && sleep 3600']
```

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# Pod Lifecycle




Resource **requests** & limits:

```
  containers:
    resources:
      requests:
       cpu: 1
       memory: "2Gi"
      limits:
       cpu: 2
       memory: "4Gi"
```

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# [Reference] Pod Detailed Status

- PodStatus.phase:
  - Pending, Running, Succeeded, Failed, Unknown
- PodConditions:
  - PodScheduled, Ready, Initialized, Unschedulable, ContainersReady

- Container states:
  - Waiting, Running, Terminated
- Container restartPolicy:
  - Always, OnFailure, Never



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# Init Containers

Good for:

- preparatory tasks,
- limiting security footprint.

Lifecycle:

- started in sequence
- retried until succeed
  - except when forbidden by restartPolicy
- resources - use max, not sum
- block standard containers

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# Common Controllers

ReplicaSet

Deployment

StatefulSet

DaemonSet

Job

CronJob

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# ReplicaSets

Keeps the desired number of controlled Pods.

- selector,
- number of replicas,
- Pod template.

**Beware** of overlapping or colliding selectors!



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# Sample ReplicaSet

```
apiVersion: apps/v1
kind: ReplicaSet
metadata:
  name: sample-replica-set
spec:
  replicas: 3
  selector:
    matchLabels:
      app: sample
  template:
    metadata:
      labels:
        app: sample
    spec:
      [...]
```

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# Deployments

Gradually **drives towards** desired state.

- controls multiple ReplicaSets
  - one for each Pod template version
  - identified via pod-template-hash label
- RollingUpdateStrategy (maxUnavailable, maxSurge)

Operations:

- rollout
- rollover
- rollback (rollout undo)
- scaling
- pause/resume

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# Sample Deployment

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: sample-deployment
spec:
  replicas: 3
  selector:
    matchLabels:
      app: sample
  template:
    metadata:
      labels:
        app: sample
    spec:
      [...]
```

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# StatefulSets

Cattle vs **pets**

Assures Pod identity

- ordinal index
- hostname: <StatefulSetName>-<Ordinal>
- storage: PersistentVolume

Sequential creation & deletion

- but podManagementPolicy=Parallel



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# Jobs & CronJobs

Designed for **batch jobs** where Pods terminate.



Job

- single execution
- fixed completion count
- work queue
- note: Pods coordinate among themselves



CronJob

- like crontab
- creates Jobs

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# DaemonSets

Pod lifetime **tied to Node**.

Running daemons for:

- distributed cluster storage,
- logging,
- monitoring,
- GPU support.

Some alternatives:

- init scripts,
- static Pods.

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# Services

Reminder: Pods are **ephemeral**:

- hostnames change,
- IP addresses change too.

**Service** object:

- an end-point for a set of Pods,
- label selector,
- port mapping(s),
- virtual IP aka clusterIP.

**Headless** Service:

- clusterIP not set,
- DNS returns A record with all matching Pods.




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# Services - cntd.

In-cluster **service discovery**:

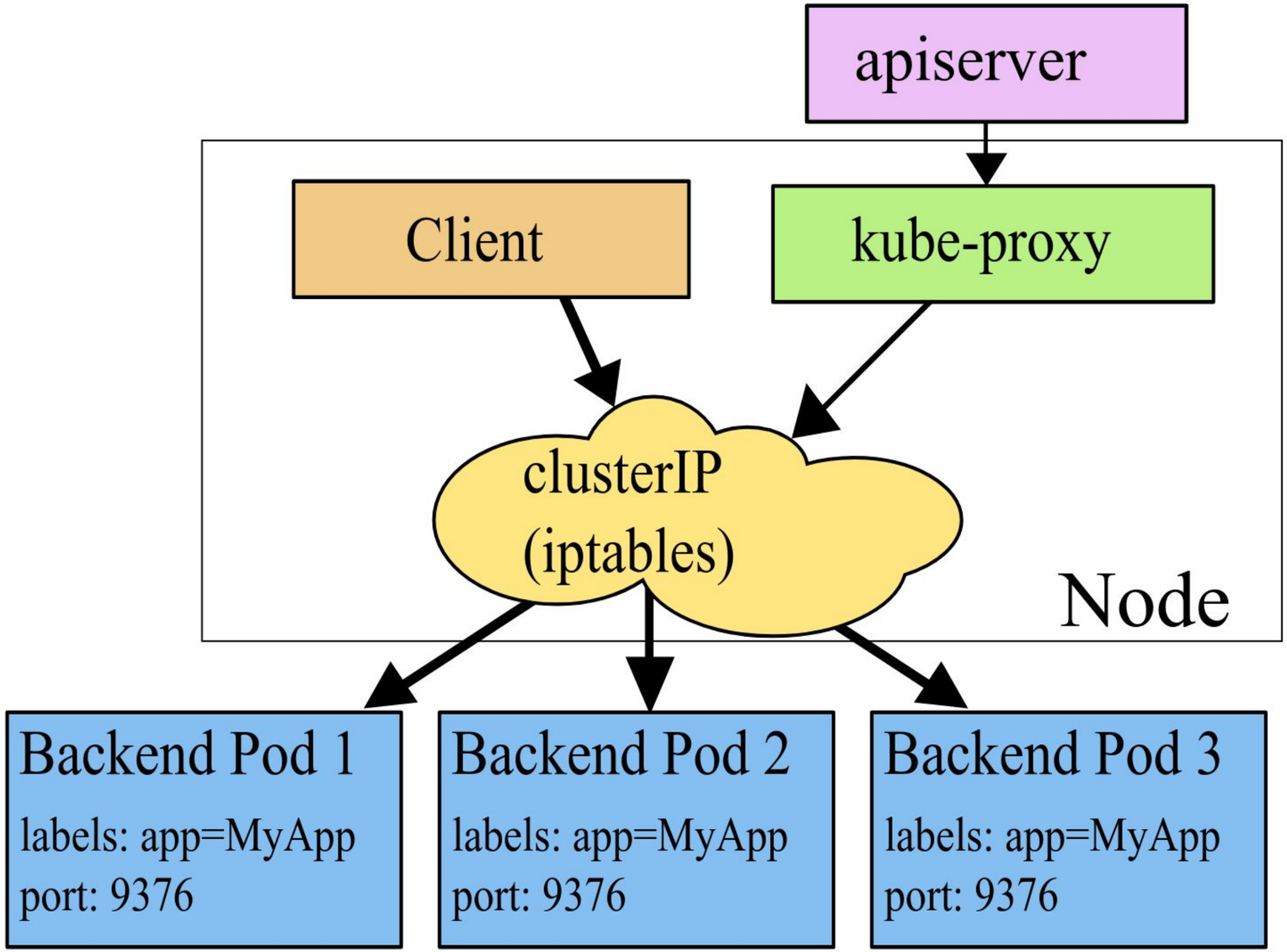
- environment variables
- **DNS**
  - service-name.namespace-name
  - service-name inside same namespace

ServiceTypes

- ClusterIP
- NodePort
- LoadBalancer
- ExternalName

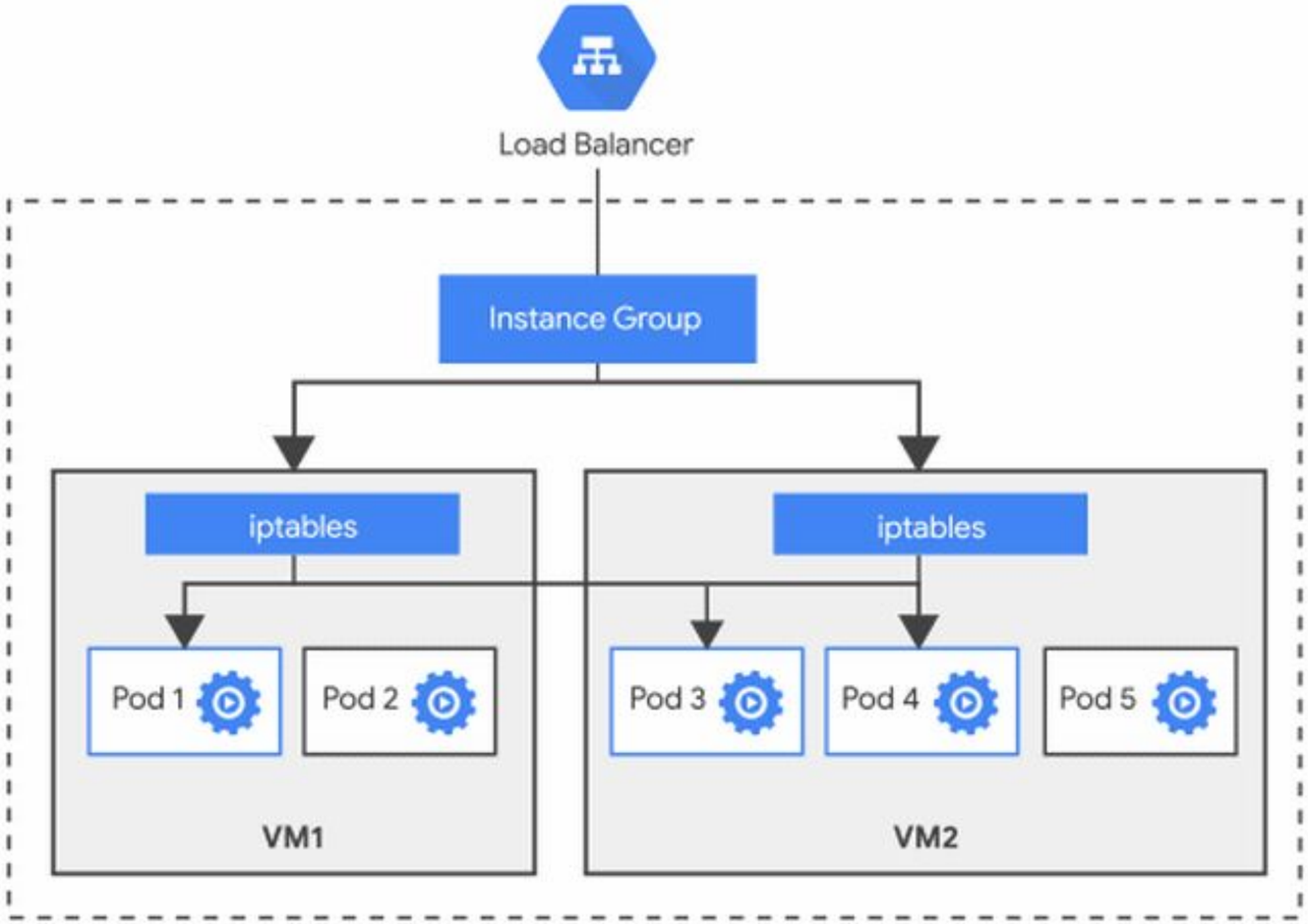
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# Magic Behind a Service




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# LoadBalancer Service




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# Storage

Reminder: Container files are **ephemeral** too.

**Volume** objects

- (usually) **independent** lifecycle
- emptyDir
  - stays as long as Pod stays on the Node
- configMap
  - ConfigMap objects as a file system
- persistentVolumeClaim for PersistentVolume
  - StorageClass, eg. **GCEPersistentDisk**
- and multiple other types...

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# PersistentVolume

PersistentVolume

- storage in the cluster, as a **resource**,
- PersistentVolumeClaim is a resource **request**,
- dynamic binding,
- static or dynamic provisioning,
- types:
  - GCEPersistentDisk
  - NFS
  - and multiple others
- access modes:
  - ReadWriteOnce
  - ReadOnlyMany
  - ReadWriteMany

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Google Cloud